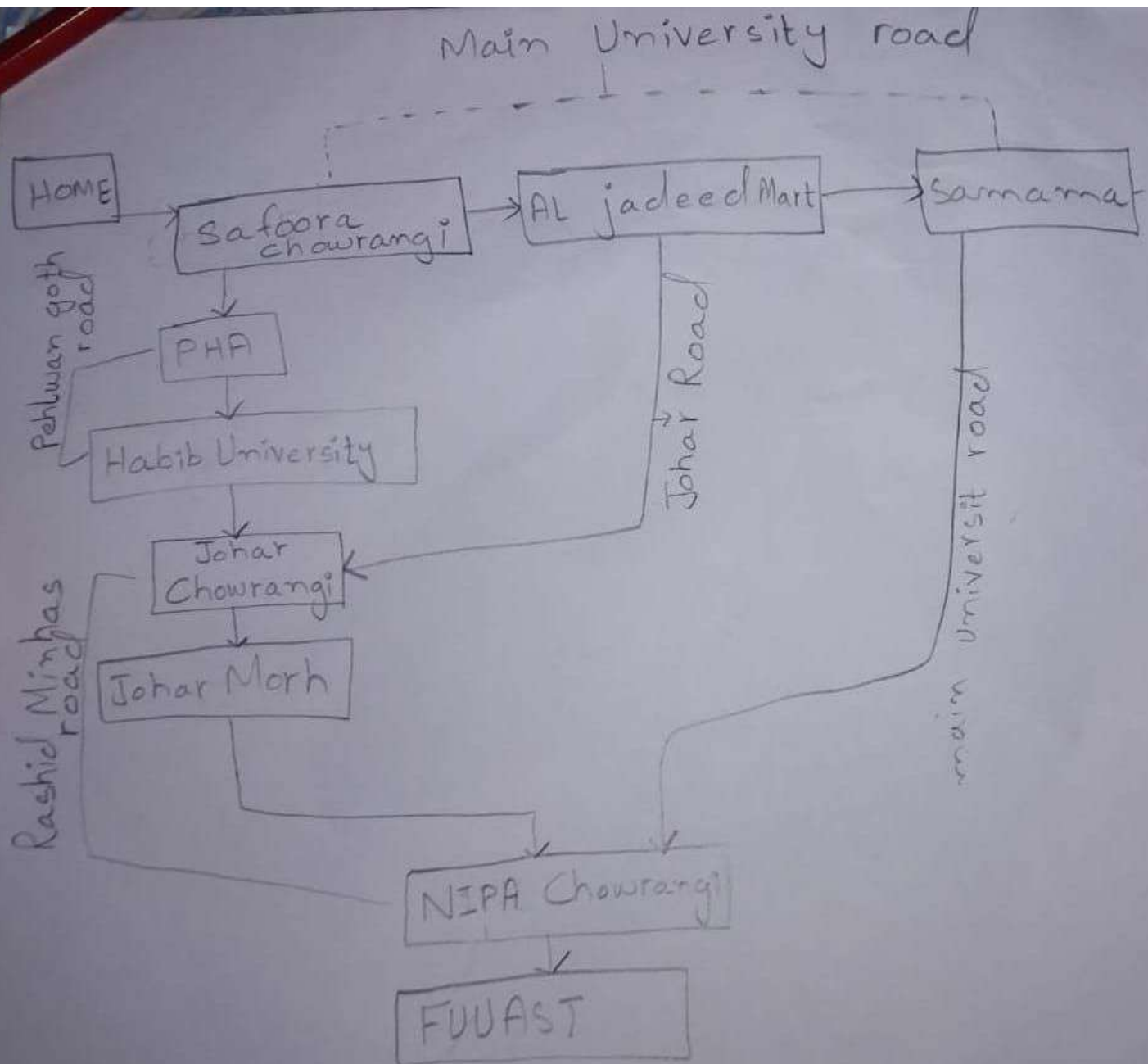
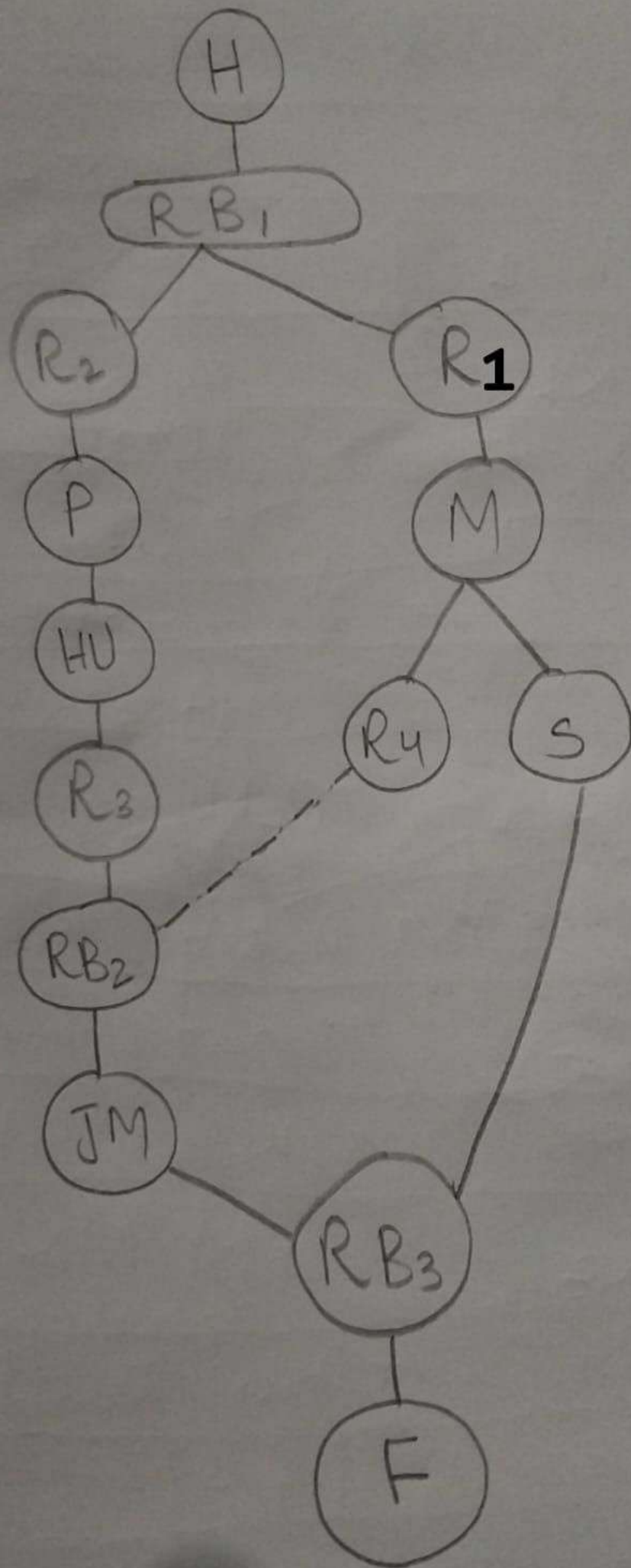




Symbol	Location
H	Home
RB ₁	Satoora Chowranggi
R ₁	Main Uni road
M	Al-jadeed Mart
S	Samama
P	PHA society
R ₂	Pehlwan goth road
HU	Habib University
RB ₂	Johar Chowranggi
JM	Johar Morh
R ₃	Rashid Minhas Road
R ₄	Johar Road
RB ₃	Nipa Chowranggi
F	FUUA ST



→ Step 0: Initialization

$$Q = [H]$$

$$V = [H]$$

→ Step 1: Dequeue H and expand

$$Q = [RB1]$$

$$V = [H, RB1]$$

→ Step 2: Dequeue RB1 and expand

$$Q = [R_2, R_1]$$

$$V = [H, RB1, R_2, R_1]$$

→ Step 3: Dequeue R_2 and expand

$$Q = [R_1, P]$$

$$V = [H, RB1, R_2, R_1, P]$$

→ Step 4: Dequeue R_1 and expand

$$Q = [P, M]$$

$$V = [H, RB1, R_2, R_1, P, M]$$

→ Step 5: Dequeue P and expand

$$Q = [M, HU]$$

$$V = [H, RB1, R_2, R_1, P, M, HU]$$

Step 6: Dequeue M and expand

$$Q = [H, U, R_4, S]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S]$$

Step 7: Dequeue HU and expand

$$Q = [R_4, S, R_3]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3]$$

Step 8: Dequeue R_4 and expand

$$Q = [S, R_3, RB_2]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2]$$

Step 9: Dequeue S and expand

$$Q = [R_3, RB_2, RB_3]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2, RB_3]$$

Step 10: Dequeue R_3 and expand

$$Q = [RB_2, RB_3]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2, RB_3]$$

Step 11: Dequeue RB_2 and expand

$$Q = [RB_3, JM]$$

$$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2, RB_3, JM]$$

Date _____

Step 12: Dequeue RB_3 and expand

$Q = [JM, F]$

$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2, RB_3, JM, F]$

Step 13: Dequeue JM and expand

$Q = [F]$

$V = [H, RB_1, R_2, R_1, P, M, HU, R_4, S, R_3, RB_2, RB_3, JM, F]$

Step 14: Dequeue F
Goal Found!

Shortest Path:

$H \rightarrow RB_1 \rightarrow R_1 \rightarrow M \rightarrow S \rightarrow RB_3 \rightarrow F$